



- Answer all Questions on your Answer Sheet (Just the answers, not your working).
- Rough paper is given for working.
- Calculators may be used (CASIO only☺).
- Diagrams are not necessarily drawn to scale.
- May the force be with you!

Hint: $(a - b)^2 = a^2 - 2ab + b^2$

5 MARK QUESTIONS

- (1) The sum of the digits of a two digit number is 6 and their product is 8.

If the number is less than 40, what number is it?

- (2) Given: $a \Psi b = \frac{a+b}{b-1}$

With $4 \Psi x = \frac{12}{7}$

Evaluate $x \Psi 7$.

- (3) Evaluate: $\sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{3 + \sqrt{1}}}}}}$

- (4) $(x - y)^2 = 39$ and $xy = -16$

So... $x^2 + y^2 = ?$

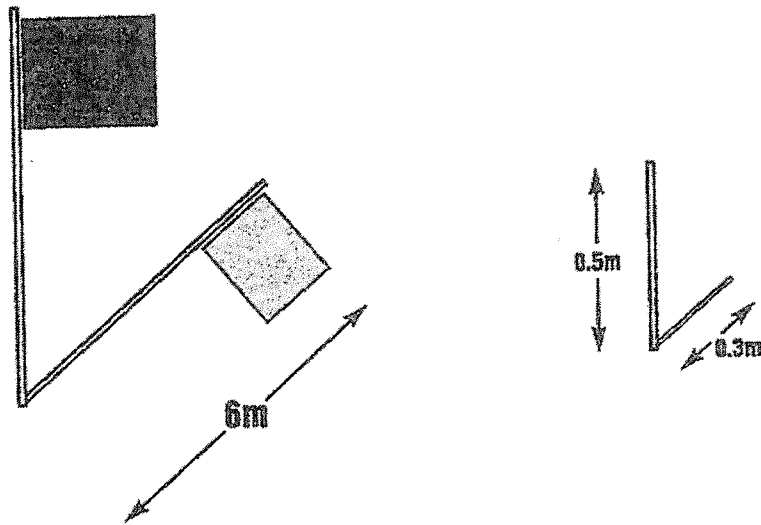
- (5) A mother is currently 30 years old. She has three children aged 5, 3 and 0. In 12 years' time, the sum of the children's ages will be half that of the mother's and the father's ages combined.

Determine how old the father is **now**.

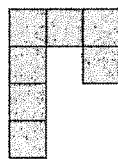


- (6) At 3pm one day, a flagpole and a measuring pole cast shadows as shown.

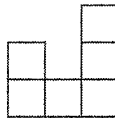
What length is the flagpole?



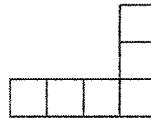
- (7) The following drawings show the top, front and side views of a three-dimensional figure.



Top view

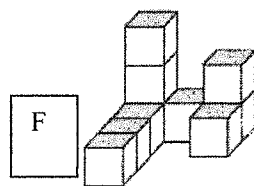


Front view

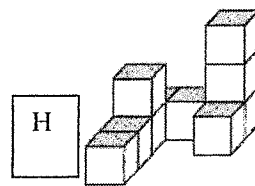


Right-side view

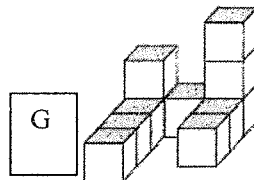
Which figure do these views represent?



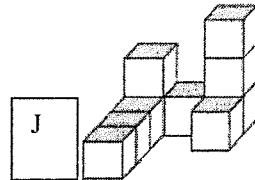
Front



Front

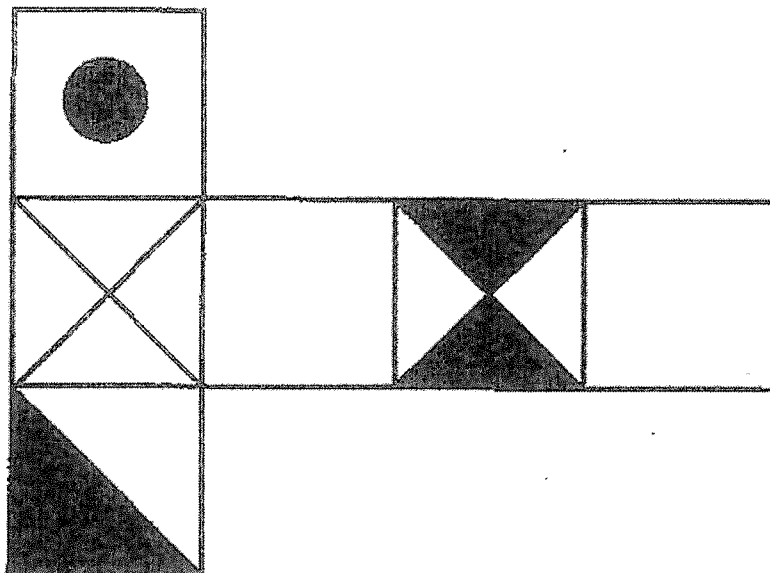


Front

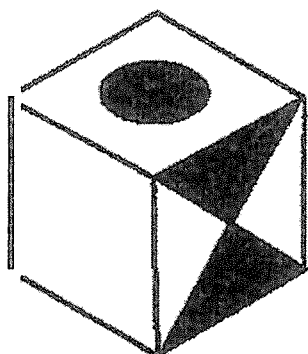


Front

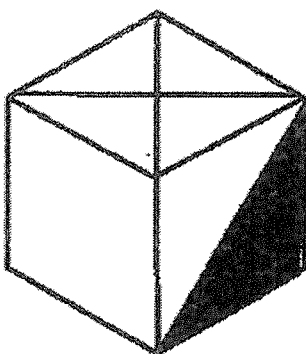
(8)



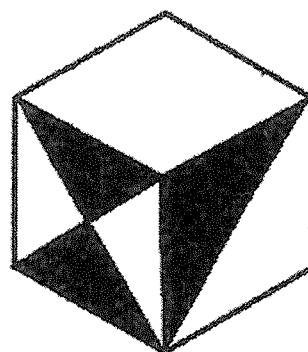
When the above is folded to form a cube, one of the figures below can be produced.
Which is it?



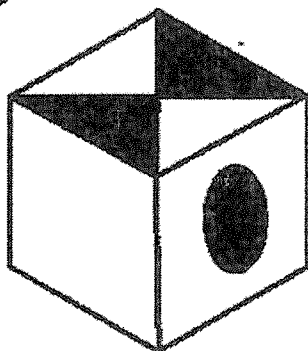
A



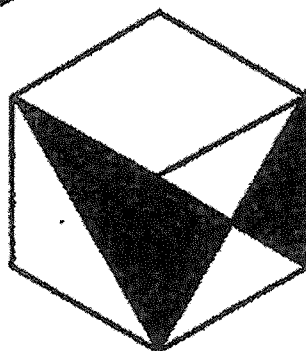
B



C



D



E

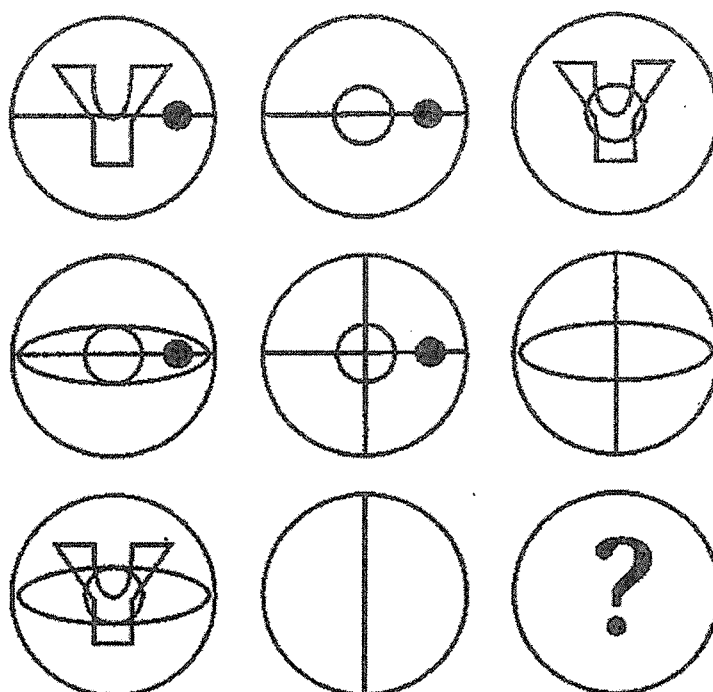
10 MARK QUESTIONS

$$(9) \quad \begin{array}{cccccccc} & & & & 1 & & & \text{Row 0} \\ & & & & & & & \\ & & & 1 & & 1 & & \text{Row 1} \\ & & & & & & & \\ & & 1 & & 2 & & 1 & \text{Row 2} \\ & & & & & & & \\ & 1 & & 3 & & 3 & & 1 & \text{Row 3} \\ & & & & & & & \\ 1 & & \cdots & & \cdots & & \cdots & 1 & \text{Row 4} \end{array}$$

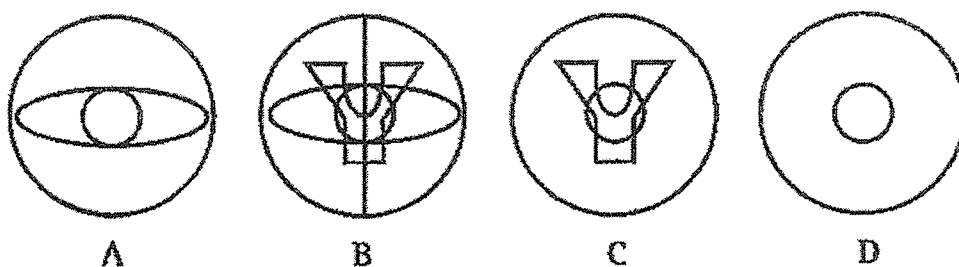
The numbers in the n^{th} row of the Pascal's Triangle represent the coefficients for $(a+b)^n$.

If a row's coefficients total 512, it is for $(a+b)^?$ *value of question mark*

(10)



Which of the circles above should replace the question mark below?



- (11) The only symbols that concern you in this multiplication puzzle are the stars. Using their positions on the grid, calculate the missing number.

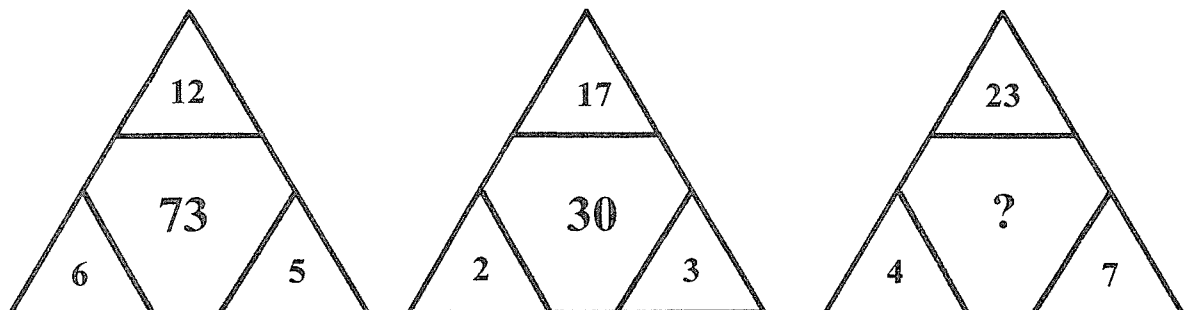
	90
	210
	48
	168
	240
	192
	72
	180
	120
	216
	?

- (12) Determine how many seams there are in a traditional-looking soccer ball.

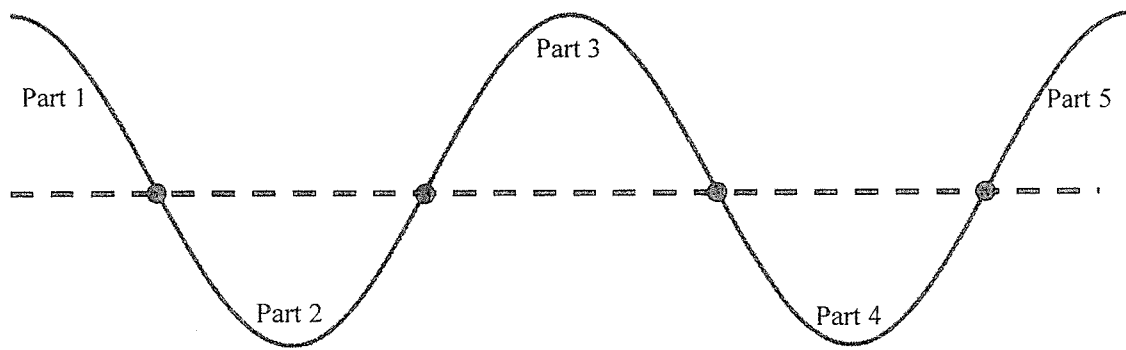
The soccer ball consists of 12 pentagons and 20 hexagons.



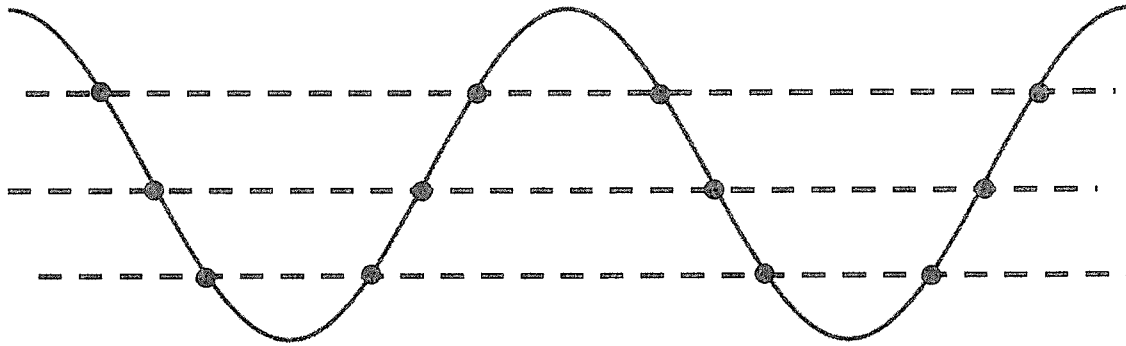
- (13) Fill in the missing number in place of the ?.



- (14) A horizontal line intersects a piece of string at four points and divides it into five different parts as show below.

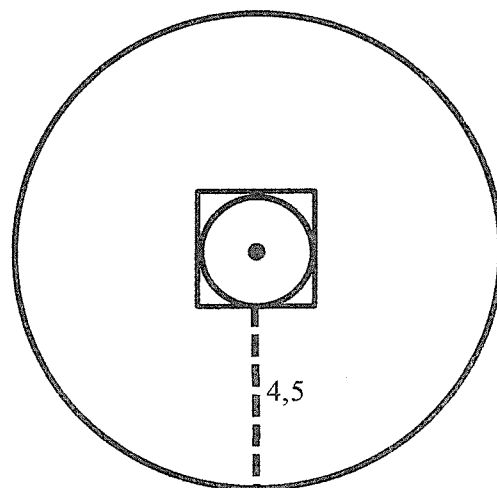


The piece of string is intersected in this way by more parallel lines, each of which intersects the string at four points.



Determine the number of parts into which the string will be divided when 2016 parallel lines intersect the string in the same way as shown above.

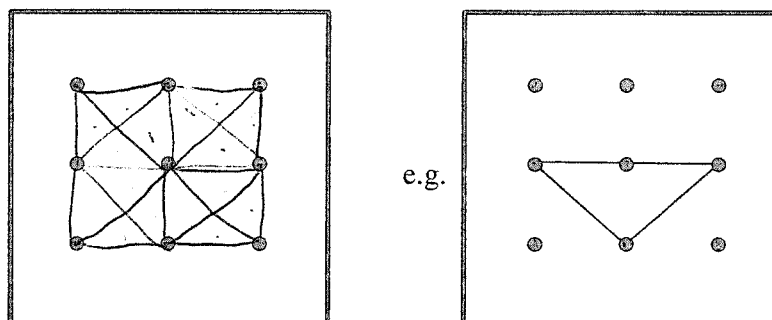
- (15) The large and small circles are concentric.



The large circle's area is 49π . Determine the area of the square.

20 MARK QUESTIONS

- (16) How many triangles can be drawn on a 3 x 3 pinboard?

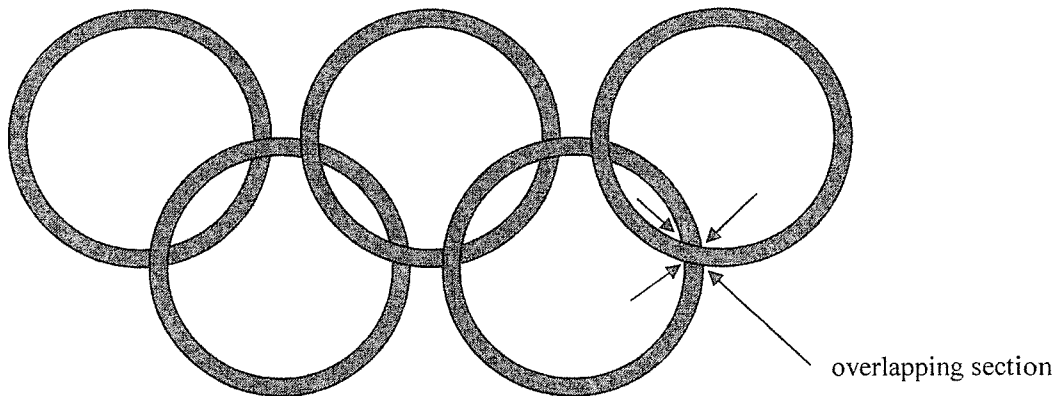


- (17) Each dot in the multiplication puzzle shown represents one of the ten digits from 0 to 9 inclusive. Some digits may appear more than once, others not at all. As you can see, a two-digit number multiplied by a two-digit number yields a four-digit product, to which is added a three digit number starting with 1. Replace each dot with the proper digit. The solution is unique.

Determine both two-digit numbers in the initial product.

$$\begin{array}{r}
 \begin{array}{cc}
 \cdot & \cdot \\
 \cdot & \cdot
 \end{array} \\
 \times \quad \begin{array}{cc}
 \cdot & \cdot \\
 \cdot & \cdot
 \end{array} \\
 \hline
 \begin{array}{cccc}
 \cdot & \cdot & \cdot & \cdot \\
 + & 1 & \cdot & \cdot \\
 \hline
 \cdot & \cdot & \cdot & \cdot & \cdot
 \end{array}
 \end{array}
 \left. \vphantom{\begin{array}{r} \cdot \\ \cdot \\ \cdot \\ \cdot \end{array}} \right\} \text{Initial product}$$

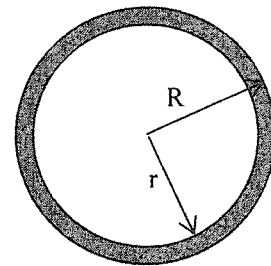
- (18) The symbol of the Olympic Games is composed of five interlocking rings. It was originally designed in 1912 by Baron Pierre de Coubertin, the founder of the modern Olympic Games.



These Olympic Rings are painted grey on a wall in Rio de Janeiro.

You are given the following information about the Olympic Rings shown above:

- All five rings are identical
- The radius, R , of the outer ring is 4 m.
- The radius, r , of the inner ring is 3,7 m.
- Each overlapping section is $0,09 \text{ m}^2$.
- The area of a circle is πr^2 .



Determine the area of the painted section of the Olympic Rings, **correct to the nearest integer**.
i.e. the shaded grey area shown on the figure.

THE END